Thrombotic thrombocytopenic purpura

Diagnosis

- MAHA (schistocytes, high LDH, low haptoglobin) + thrombocytopenia ± neuro, renal, fever
- **ADAMTS13 < 10%**; PLASMIC score ≥ 6 high probability

Emergency management

- **Plasma exchange daily** (start before ADAMTS13 result)
- Steroids, **caplacizumab**, rituximab; avoid platelet transfusion unless bleeding

TMA differential

- HUS (STEC/atypical), DIC, HELLP, malignant hypertension, drug-induced, scleroderma crisis

Key point: Suspected TTP is an emergency: PLEX without delay.

Haemolytic uraemic syndrome

Classification

- STEC-HUS (E. coli O157), atypical (complement-mediated), secondary (pregnancy, drugs, transplant)

Atypical HUS

- **Eculizumab** (anti-C5) after excluding TTP; meningococcal vaccination
- Plasma exchange as bridge

STEC-HUS

- Supportive, RRT; avoid antibiotics and antimotility drugs

Key point: Complement blockade transforms aHUS outcome.

Heparin-induced thrombocytopenia

4T score

- Thrombocytopenia (> 50% fall), timing (5–10 days), thrombosis, no other cause
- ≥ 4 intermediate/high → stop heparin, test

Pathophysiology

- IgG against PF4–heparin complexes → platelet activation and thrombosis

Management

- **Stop all heparin**; argatroban or bivalirudin (or fondaparinux)
- Anti-PF4 ELISA, functional assay (SRA); no platelet transfusion
- Warfarin only when platelets recover (> 150)

Key point: HIT causes thrombosis, not bleeding: anticoagulate with a non-heparin agent.

Disseminated intravascular coagulation

ISTH overt DIC score

- Platelets, fibrin markers (D-dimer), PT prolongation, fibrinogen; ≥ 5 = overt DIC

Pathophysiology

- Tissue factor-driven thrombin generation, consumption of factors, fibrinolysis inhibition (PAI-1) in sepsis

Management

- **Treat the cause** (source control, antibiotics)
- Platelets < 50 if bleeding (< 10–20 if not), FFP, cryoprecipitate for fibrinogen < 1.5
- Prophylactic heparin if not bleeding

Key point: DIC is a sign of disease: treat the cause, replace if bleeding.

Thrombocytopenia in ICU

Causes

- Sepsis, DIC, dilution, drugs (heparin, linezolid, beta-lactams), HIT, TMA, ECMO, liver disease

Diagnostic approach

- Blood film (schistocytes, pseudothrombocytopenia), coagulation, 4T, ADAMTS13

Prophylactic thresholds

- < 10 ×10^9/L no bleeding; < 20 CVC (ultrasound guided: PACER — avoid transfusion)
- < 50 surgery/LP; < 100 neurosurgery

Key point: Check the film before platelets: exclude HIT and TTP.

Massive transfusion and ROTEM

Massive transfusion

- ≥ 10 units RBC in 24 h or ≥ 4 in 1 h
- **Damage control**: permissive hypotension (not in TBI), **1:1:1** (PROPPR), TXA within 3 h (CRASH-2)

ROTEM-guided therapy

- A5 FIBTEM < 10 mm: fibrinogen/cryo
- EXTEM CT > 80 s: FFP/PCC; EXTEM A5 < 35 + FIBTEM normal: platelets
- ML > 15%: TXA

Complications

- Hypocalcaemia, hyperkalaemia, hypothermia, acidosis, TRALI, TACO

Key point: Viscoelastic testing guides targeted correction.

TRALI and TACO

Differentiate

- TRALI: within 6 h, non-cardiogenic oedema, fever, hypotension, normal BNP
- TACO: hypertension, high BNP, positive balance, response to diuretics

Management

- TRALI: stop, supportive, ventilation, no diuretics
- TACO: stop, diuretics, NIV

Other reactions

- Febrile, allergic, anaphylaxis (IgA), haemolytic (ABO), septic

Key point: TRALI is low pressure; TACO is high pressure.

Complications after stem-cell transplantation

By phase

- Pre-engraftment (< 30 days): neutropenic sepsis, mucositis, VOD/SOS, engraftment syndrome
- Early (30–100): acute GVHD, CMV, fungal infection
- Late (> 100): chronic GVHD, encapsulated bacteria

Differential

- Acute GVHD (skin, gut, liver), engraftment syndrome, infection, drug reactions, VOD

Management

- **High-dose methylprednisolone** for GVHD; ruxolitinib if refractory
- Defibrotide for VOD; antimicrobials

Key point: Rash + diarrhoea + jaundice after HSCT → acute GVHD.

Haemophagocytic lymphohistiocytosis

Definition and triggers

- Hyperinflammation from uncontrolled macrophage activation
- Triggers: viral (EBV), malignancy (lymphoma), autoimmune (MAS)

Diagnosis

- **HLH-2004** (5 of 8: fever, splenomegaly, cytopenias, high TG/low fibrinogen, ferritin > 500, sCD25, NK activity, haemophagocytosis)
- **HScore**; ferritin often > 10,000

Management

- Dexamethasone, etoposide, anakinra, IVIG; treat trigger (rituximab for EBV)

Key point: Very high ferritin + cytopenias + fever → think HLH.

Anaemia and transfusion thresholds

Causes

- Blood sampling, bleeding, inflammation (hepcidin), renal failure, haemolysis, nutrition

Thresholds

- **Hb 7 g/dL** (TRICC, TRISS); 8 if cardiac (MINT 2023 suggests ≥ 10 in MI)

Reducing anaemia

- Less sampling (small tubes), cell salvage, IV iron, EPO (selected)

Key point: Restrictive transfusion is safe in most ICU patients.

Anticoagulant reversal

Warfarin reversal

- **4-factor PCC 25–50 IU/kg + vitamin K 5–10 mg IV**
- FFP if PCC unavailable

DOAC reversal

- Dabigatran: **idarucizumab 5 g**
- Xa inhibitors: andexanet alfa or PCC 50 IU/kg

Restarting

- Mechanical valve: restart heparin after 48–72 h in ICH (balance risk)
- AF: 7–14 days depending on bleed severity

Key point: PCC is faster and better than FFP for warfarin reversal.

Sickle cell crisis and acute chest syndrome

Acute chest syndrome

- New infiltrate + chest pain, fever, hypoxia; causes: infection, fat embolism, infarction

Management

- O2, analgesia, incentive spirometry, antibiotics (macrolide + cephalosporin)
- **Transfusion**: simple or exchange (HbS < 30%)

Other complications

- Vaso-occlusive crisis, stroke, splenic sequestration, aplastic crisis, priapism

Key point: Exchange transfusion for severe ACS.

Guidelines (no 2026 note on this topic)